ASEN 4018-013,014

SENIOR PROJECTS

An Autonomous and Cooperative UAS Multirotor and Rover Network for Exploration, Search, and Payload Delivery/Retrieval

FALL 2026



PROJECT INTRODUCTION

Melvin Rafi

Smead Aerospace
UNIVERSITY OF COLORADO BOULDER



WELCOME TO ASEN 4018-013,014 SENIOR PROJECTS LAB!

Instructional Team

Instructor



Dr Melvin Rafi
Office: AERO N203
Email: melvin.rafi@colorado.edu

Teaching Facilitators

Jacob Nelson (Section 013, TTh 10:40am-12:30pm)
Email: Jacob.Nelson-1@colorado.edu

Kirin Kawamoto (Section 014, TTh 1:00pm-2:50pm)
Email: Kirin.Kawamoto@colorado.edu

Modes of Contact

Lab Time & Office Hours – for guidance and questions on the project, concept and knowledge building, etc.

MS Teams – centralized communication and collaboration platform for the entire team.

Email – for other questions or concerns regarding individual matters, etc.



PROJECT BACKGROUND AND REQUIREMENTS

Project Background

An Autonomous and Cooperative UAS Multirotor and Rover Network for Exploration, Search, and Payload Delivery/Retrieval

What is the big picture idea behind this project?

- In search and exploration missions, it may sometimes be advantageous to utilize a ***multi-agent network*** of air and land-based vehicles to ***explore an area of interest and to deliver or retrieve payloads*** at locations with reduced accessibility, where the precise location may not be initially known.
- Each vehicle type has advantages within its own operating environment. A UAS has the advantage of ***maneuvering over terrain to perform a wider area search***, but may not have the ability to land in areas lacking sufficient clearance. In such cases, the rover has the advantage of ***maneuvering on terrain inaccessible by the UAS***.
- Within this multi-agent network, a UAS multirotor may take on the roles of search, exploration, and determination of a precise location of interest (for example, a stranded hiker), while a rover may take on the roles of payload delivery or retrieval once the precise location has been determined by the UAS.
- Each vehicle ***autonomously and cooperatively works together*** to increase the overall chances of mission success.

Mission Objective

The objective of the UAS multirotor and rover network is to autonomously and cooperatively explore an area of interest to locate and deliver/retrieve a payload at a desired location of interest.

This project involves developing and prototyping a network comprising of one UAS and one rover that can function autonomously (or semi-autonomously) and cooperatively. A successful vehicle network should be able to explore an area of interest, determine a precise location for payload delivery/retrieval, and deliver/retrieve a payload at the desired location.

Mission Profile

1) **UAS is given coordinates of an area of interest and takes off from starting point A** (manual control).

2) **UAS flies to the area of interest** (autonomous/semi-autonomous with manual override)

3) **UAS starts a search within the area of interest to detect an ArUco marker** (autonomous/semi-autonomous with manual override)

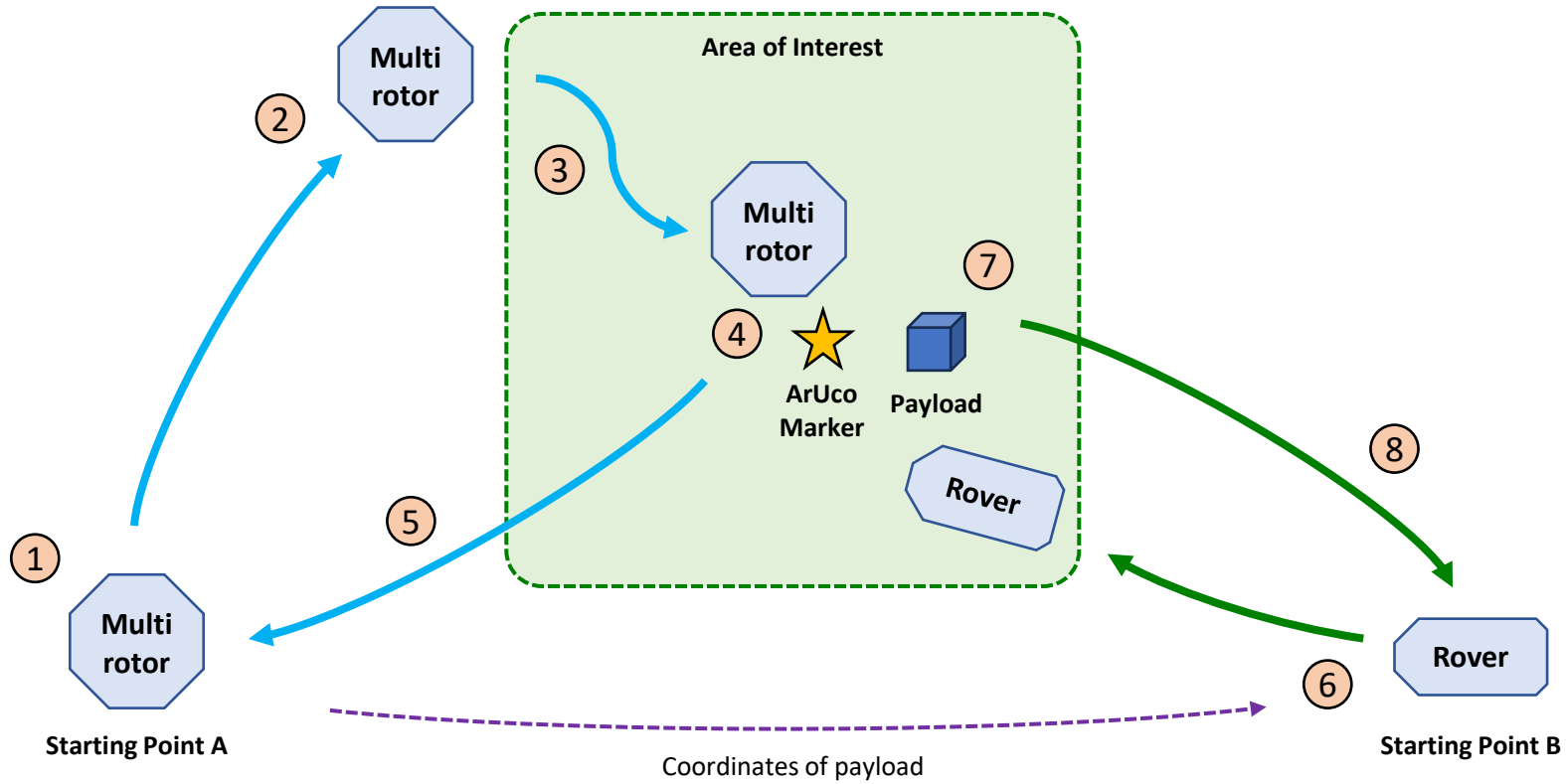
4) **UAS detects marker and relays marker coordinates to the rover or base station**

5) **UAS returns to its starting location** (autonomous/semi-autonomous with manual override)

6) **Rover receives coordinates of marker and moves from starting point B towards marker** (autonomous/semi-autonomous with manual override)

7) **Rover delivers/retrieves payload within a radius of the marker** (manual control)

8) **Rover returns to its starting location.** (autonomous/semi-autonomous with manual override)



Note: This purpose of this illustration is to give your team a ***basic idea*** of the concept, and not to "bound" or "define" your team's Concept of Operations or solution. ***As a team, you should arrive at your own CONOPS and solution to the problem***.

Initial Requirements

General Requirements

1) The multi-agent network should comprise one UAS and one rover working autonomously/semi-autonomously and cooperatively to search an area of interest, determine the coordinates of a location of interest, and deliver/retrieve a payload at the location of interest, which is specified by an ArUco marker.

Initial Requirements

UAS Team

1) The UAS multirotor platform should be able to contain all necessary hardware, sensors, communications equipment, and miscellaneous electronics, as required, to support safe fulfilment of the mission objectives.

2) The UAS shall have a volumetric footprint of at most 50 cm length x 50 cm width x 30 cm height, shall have an endurance of at least 15 minutes, shall be able to function and operate in temperatures ranging from 15°C to 25°C, and shall weigh no more than 25 lb (11.3 kg).

3) The UAS shall be able to takeoff from a starting location, autonomously maneuver to a predetermined area of interest, autonomously search the area of interest to detect an ArUco marker to within a 30 cm radius of accuracy, relay the coordinates of the marker to the rover or a base station, and return to its starting location.

4) For testing and demonstration of the prototype, the UAS shall maneuver at an altitude no higher than 2m above ground level, at a speed of no more than 0.5 m/s, and must remain within visual line of sight (the team should ensure that these parameters are in compliance with and approved by CU Flight Ops).

5) The autonomous phases of flight shall be carried out semi-autonomously where necessary or appropriate. Each autonomous phase of flight must have an override for manual control.

6) The UAS must comply with all relevant FAA regulations and CU Flight Ops policies.

Initial Requirements

Rover Team

1) The rover platform should be able to contain all necessary hardware, sensors, communications equipment, and miscellaneous electronics, as required, to support safe fulfilment of the mission objectives.

2) The rover shall have a volumetric footprint of at most 30 cm length x 30 cm width x 30 cm height (excluding the payload delivery/retrieval mechanism), shall have an endurance of at least 60 minutes, and shall be able to function and operate in temperatures ranging from 15°C to 25°C.

3) The rover shall be able to receive coordinates of the ArUco marker from the UAS or a base station, autonomously move from a starting location to the marker location, deliver/retrieve a payload within a 30 cm radius of the marker location, and return to its starting location. Payload delivery/retrieval may be conducted via manual control of the rover.

4) The rover shall be able to navigate over terrain features of 3cm height without having its movement impeded.

5) The autonomous phases of motion shall be carried out semi-autonomously where necessary or appropriate. Each autonomous phase of motion must have an override for manual control.

Initial Requirements

Definitions

1) ‘Area of interest’ is defined for testing purposes as a space of approximately 5m x 5m. Testing may be conducted indoors or outdoors, depending on availability of labs or outdoor spaces. All testing must meet the necessary safety requirements and approvals before testing can commence.

2) ‘Location of interest’ is defined as a singular coordinate where an ArUco marker is located.

3) ‘Payload’ is defined as an object of size 8cm x 8cm x 8cm and mass of no more than 1kg.

4) ‘Terrain feature’ is defined as an object of at least 3 cm height.

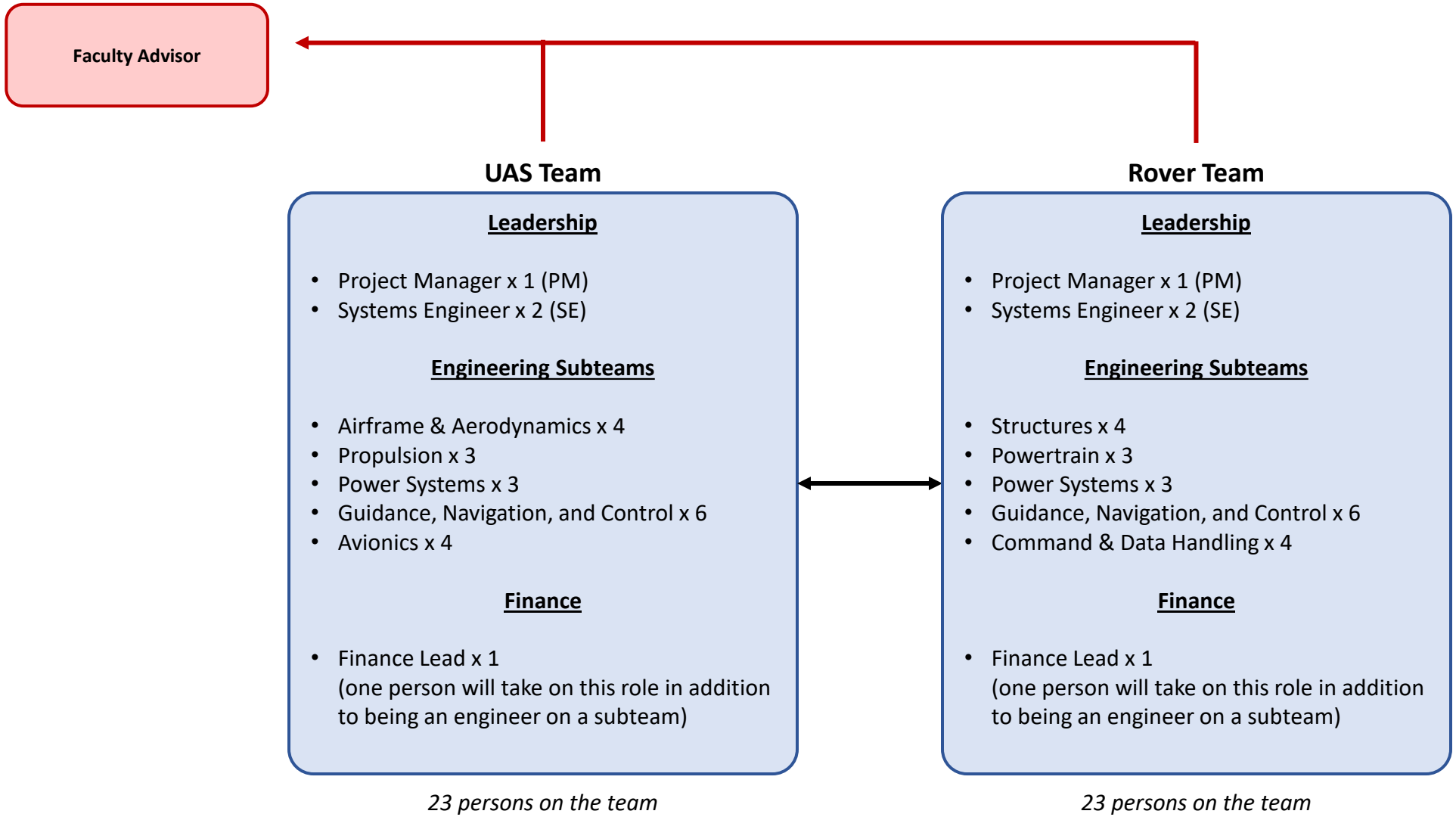
Additional Notes

1) Section 013 will focus on the payload delivery requirement for the rover, while Section 014 will focus on the payload retrieval requirement for the rover.

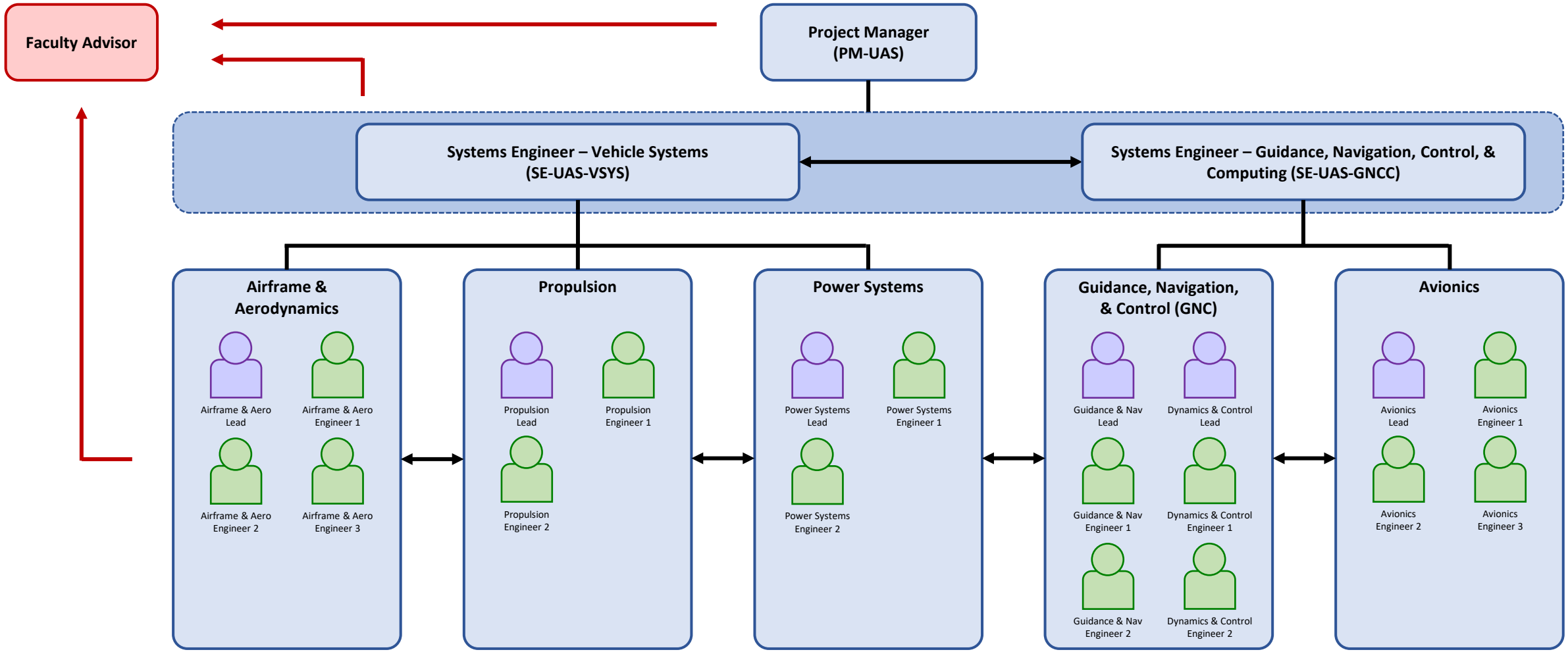


TEAM STRUCTURE

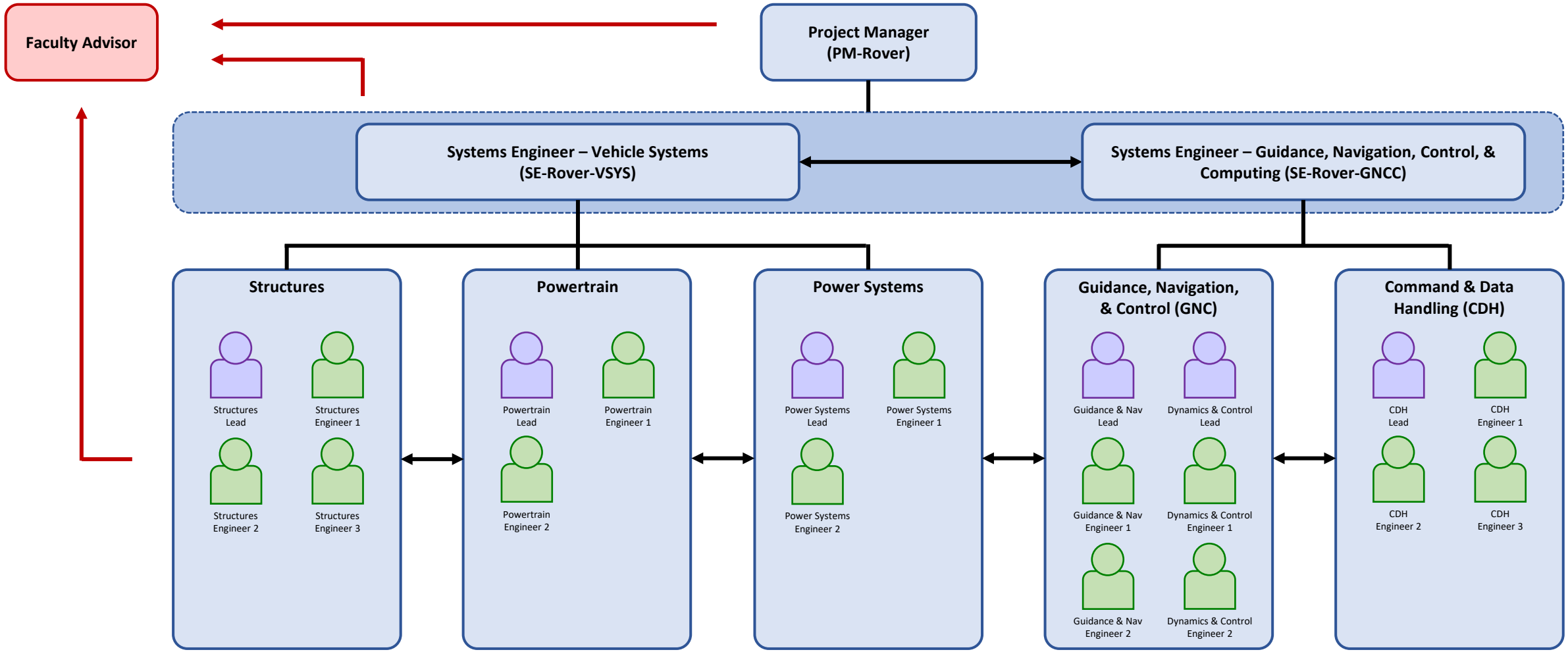
Team Structure Within Each Section



Team Structure – UAS



Team Structure – Rover



Leadership Roles

Role	UAS / Rover
Project Manager **(PM-UAS and PM-Rover)**	Responsible for overall project execution, including schedule, resources, milestones, risk, budget, team coordination, and communication with faculty and relevant project stakeholders. Ensures that the team maintains alignment with mission objectives and systems engineering processes while coordinating with systems engineers and across subsystem teams. ***Are we going to deliver the project?***
Systems Engineer – Vehicle Systems **(SE-UAS-VSYS and SE-Rover-VSYS)**	Responsible for the technical architecture and integration of the vehicle's physical systems, including airframe/structures, power, and propulsion/powertrain. Leads subsystems requirements and interfaces, evaluates system-level trades together with subsystem leads, and ensures that the vehicle's physical systems collectively satisfy mission objectives, performance goals, and functional/design requirements. ***Is the physical vehicle architected correctly?***
Systems Engineer – Guidance, Navigation, Control, & Computing **(SE-UAS-GNCC and SE-Rover-GNCC)**	Responsible for the technical architecture and integration of the vehicle's guidance, navigation, control, and computing systems. Leads subsystem requirements and interfaces, evaluates system-level trades together with subsystem leads, and ensures that sensing, estimation, control, communications, and computation work together as an integrated system. Ensures that the GCC systems collectively satisfy mission objectives, performance goals, and functional/design requirements. ***Are the vehicle's control and computing systems architected correctly?***

Engineering Roles

Subteam Function	UAS	Rover
Primary Vehicle Structure	**Airframe & Aerodynamics** Responsible for the structural design, aerodynamic performance, and physical configuration of the flight vehicle. Owns airframe sizing, structural integrity, aerodynamics, weight & balance, and mechanical integration of vehicle components.	**Structures** Responsible for the structural design, architecture, mechanical integrity, and physical integration of the rover. Owns the chassis, structural components, mechanical interfaces, load path analysis, and vehicle subsystems packaging.
Propulsion / Locomotion	**Propulsion** Responsible for converting electrical power into thrust and ensuring the flight vehicle achieves its required performance. Owns motors, electronic speed controllers, propellers, propulsion sizing, and propulsion system characterization and integration.	**Powertrain** Responsible for converting electrical power into mechanical power and delivering it to the rover's mobility systems. Owns motors, motor controllers, gearing & transmission systems, drivetrain components, and traction system characterization and integration.
Energy Systems	**Power Systems** Responsible for generating, storing, distributing, and managing electrical power systems on the flight vehicle. Owns batteries, power distribution, power conversion, power safety aspects, and power system monitoring.	**Power Systems** Responsible for generating, storing, distributing, and managing electrical power systems on the rover. Owns batteries, power distribution, power conversion, power safety aspects, and power system monitoring.
Vehicle Control	**Guidance, Navigation, & Control** Responsible for characterizing the dynamical behavior of the flight vehicle and the algorithms and systems that measure or estimate vehicle state, plan motion, and control the flight vehicle. Owns guidance, navigation, state estimation, flight-control laws, and open- and closed-loop performance.	**Guidance, Navigation, & Control** Responsible for characterizing the dynamical behavior of the rover and the algorithms and systems that measure or estimate vehicle state, plan motion, and control the rover. Owns localization, navigation, path planning, trajectory generation, control laws, and open- and closed-loop performance.
Computing & Electronics	**Avionics** Responsible for the flight vehicle's onboard electronics, computing, sensing, and communications infrastructure. Owns flight computers, sensors, data buses, telemetry, communications, software interfaces, and integration of the electronic systems supporting vehicle operations.	**Command & Data Handling** Responsible for the rover's onboard electronics, computing, sensing, and communications infrastructure. Owns vehicle computers, sensors, data buses, telemetry, communications, software interfaces, and integration of the electronic systems supporting vehicle operations.

Leadership Role Assignments

Individuals interested in leadership roles will be asked to email the Instructional Team a Leadership Statement describing why they are interested in the role, how they plan to effectively lead the team in the role, and their relevant experience that makes them suited for the role. Maximum of 500 words – do not exceed.

Timeline

- **Tuesday, Aug 25, by 11:59pm** – Individuals interested in PM/SE roles must email their Leadership Statement to the Instructional Team by this time (Please note that late submissions will not be accepted).
  - Email subjects should be titled "ASEN 4018-[Section #] – Leadership Statement for [Role Name]".
  - The candidate must indicate the specific role they are interested in – eg. "I am interested in the PM role for the Rover team".
- **Wednesday, Aug 26, by 5:00pm** – Potential candidates selected for the PM/SE roles will be notified via email. Potential Candidates will be asked to give a short 5-minute pitch to the section during lab on Thursday, Aug 27.
- **Thursday, Aug 27** – Potential candidates will give a short 5-minute pitch to the section during lab. The section will then vote on who they want as PM/SE via a Canvas survey.
- **Tuesday, Sep 1, AM** – Leadership role assignments will be announced before lab.

Engineering Role Assignments

All individuals in the section (including those who have submitted Leadership Statements) will be asked to complete a survey to rank their preference to be assigned to either the UAS team or Rover team, and to rank their preference to be assigned to one of the five engineering subteams.

Timeline

- **Tuesday, Aug 25, PM** – Engineering role surveys will open after lab. Instructions on how to complete the survey will be provided.
- **Thursday, Aug 27, at 5:00pm** – Engineering role surveys will close.
- **Tuesday, Sep 1, AM** – Engineering role assignments will be announced before lab.

We will do our best to assign each individual to a role that meets their preference, but please note that this is ***not guaranteed****.*

Next Steps

1) Review the RFP, Project Introduction slides, and initial requirements – ask me questions to clarify these objectives and requirements.

2) Submit Leadership Statement (if interested) and engineering role survey (everyone) by the respective due dates.



LAB FORMAT

Typical Weekly Schedule

Mondays

Lectures
(as per Course Schedule)

Tuesdays

Leadership team and Engineering subteams
provide updates to Faculty Advisor and entire team
(approximately 20-30 minutes)

Faculty Advisor check-ins with Leadership Team
(10 minutes)

Faculty Advisor check-ins with subteams
(approximately 20-30 minutes)

Office Hours +
Working time for team to make progress on project

Attendance is required at labs

Wednesdays

Seminars
(as per Course Schedule)

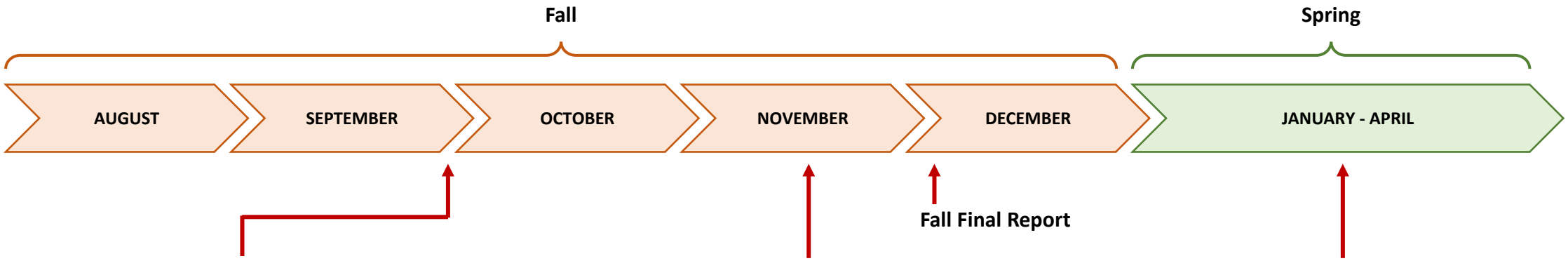
Thursdays

Leadership team and Engineering subteams
brief discussions with Faculty Advisor as required

Working time for team to make progress on project

Attendance is required at labs

Looking Ahead



Design Review 1

CONOPS + Functional Block Diagrams, Initial Modeling + Analyses + Trade Studies, Current Design Space, Modeling and Prototyping Plans, etc.

"How can we use the requirements to inform the initial models and trade studies, and how can we use the results of the trade studies to then inform the solution and design of the prototype?"

Design Review 2

Design Overview of Final Prototype, Analytical or Computational models that predict performance and demonstrate that requirements will be met, etc.

"This is the design of our final prototype and here are the engineering models that show that the prototype will work and meet the requirements."

Design Reviews, Reports, etc.

Test Plans and Safety Reviews, etc.

"Now that we've built our prototype, how are we going to test it so that we can get data to show that the design meets requirements."

Testing of Prototype, Test Results, Verification and Validation Conclusions, Program Management Lessons, etc.

"Now that we've tested our prototype, does the actual performance data validate that our models from the Fall were accurate? Does the data show that the requirements from the Fall were met?

Industry Mentors

Industry mentors can give advice on program management/schedules/risk, systems engineering/requirements, technical topics, etc. Industry mentors can also usually meet on a regular or semi-regular basis with the team and/or attend design reviews.

First Steps

- Industry Mentors will be announced soon.
- The team should assign a student to be a prime contact that liaises between the team and the industry mentor.
- The team should set up a kick-off meeting with the industry mentor to introduce themselves, explain the project mission and background, introduce the CONOPS, and get feedback. They should also seek out their mentor's availability for regular meetings thereafter.

Next Steps

1) As a team, start working on your Concept of Operations (CONOPS) and requirement flow down. You can start on this before team role assignments are announced.

2) Submit Engineering Role Surveys by today (Aug 27) at 5pm, and submit Leadership Role Surveys by tomorrow (Aug 28) at 5pm.

3) Once team role assignments are announced, meet within your respective subteams and select a person to be the subteam lead.